Data Collection and Preprocessing Phase

Date: 10 Feb 2026
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Project Title: Diabetic Retinopathy Classification Web Application
Maximum Marks: 2 Marks

Data Collection Plan

Project Overview

This deep learning project focuses on classifying retinal fundus images into different stages of Diabetic Retinopathy using Convolutional Neural Networks (CNNs).

The objective is to automatically detect the severity level of Diabetic Retinopathy from eye images and assist in early diagnosis.

The classification categories include:

- No DR
- Mild
- Moderate
- Severe
- Proliferative DR

Data Collection Plan

The dataset has been sourced from publicly available medical image repositories such as Kaggle Diabetic Retinopathy datasets.

The dataset contains labeled retinal fundus images categorized into different severity levels of Diabetic Retinopathy.

The images are stored in structured subdirectories based on classification labels.

Raw Data Sources Identified

The raw data includes labeled retinal images downloaded from Kaggle and other open medical datasets.

These images are used for training, validation, and testing of the deep learning model.

Raw Data Sources

Source Name	Description	Location/URL	Format	Size	Access Permissions
Kaggle – Diabetic Retinopathy Dataset	Publicly available labeled retinal fundus images	https://www.kaggle.com/	JPG/PNG	~ 250 MB	Public
Local Training Dataset	Processed and organized retinal images stored locally	Local Storage	JPG/PNG	~ 200 MB	Private
Sample Test Images	Sample retinal images used for testing the web app	Local Storage	JPG	~ 20 MB	Private